# Lecture IX - Safety Planning for the Islamic Pilgrimage

Applied Optimization with Julia

Dr. Tobias Vlćek

## Introduction

### Islamic Pilgrimage

**Question:** Have you ever heard of the Hajj?

### The Hajj

- **The great Islamic pilgrimage towards Mecca**
- The holy city is the **religious center** of the Islamic religion
- Located in the **Kingdom of Saudi Arabia**
- Each Muslim who is able should **perform Hajj once**
- **Confined spaces** around the holy sites
- Only **a few million** people are allowed annually

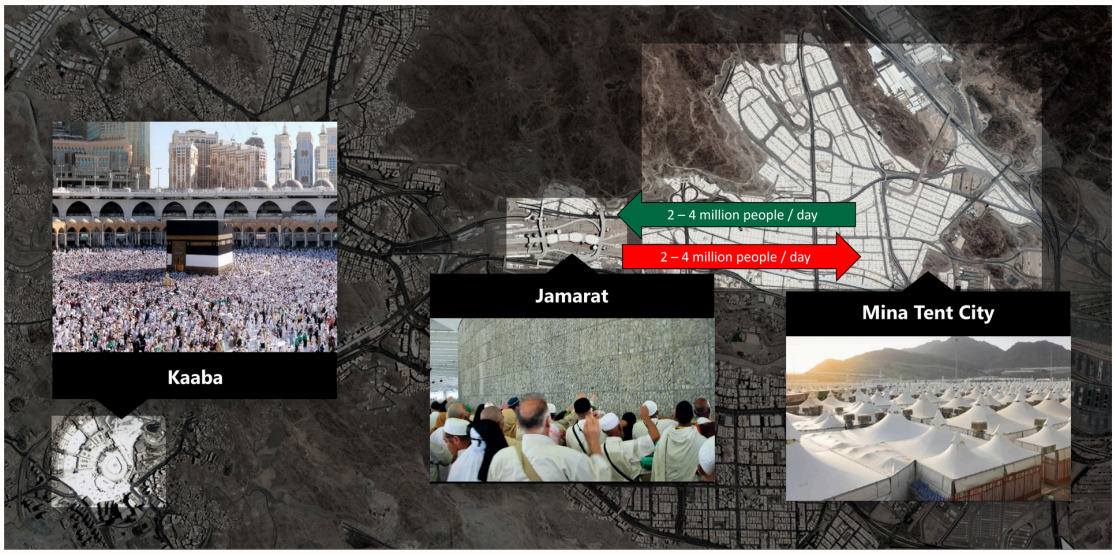
### Learning Goals

After this lecture, you will be able to:

- Structure a real-world **scheduling problem**
- Use **subsets** to keep large models small
- Model time preferences with **precomputed penalties**
- Limit **fluctuations** between consecutive periods

## Mina Tent City

## Mina Tent City




## The scope of the Hajj

- Pilgrimage is actually a **multi-day journey**
- Involves a number of **different rituals at several ritual sites**
- Our efforts focused on the **Ramy al-Jamarat ritual**

> **Ramy al-Jamarat ritual**
> Pilgrims throw pebbles against three pillars, which symbolize the temptations of the devil. They repeat this ritual with small variations on four consecutive days.

## Mina Tent City

- 1.5–2 million people reside in the tent city
- Pilgrims repeatedly access the holy site
- Perform the Ramy al-Jamarat ritual
- Walk through a network of streets and pathways
- Later proceed to the Kaaba or return to the camp

## Time Preferences

- **When to perform the ritual on each of the four days?**
- Pilgrims have different **time preferences**
  - Constrained by **arrival and departure** shuttle times
  - Extremely **hot at midday** quickly leading to exhaustion
  - Traditions play an **important role** in time preferences

**Question: What could become a problem?**

## Risk: Overcrowding at Mina

- In aggregation, the time preferences are **clustered**
- **Popular peak times** dating back to the Prophet Mohammad
- Equates to a **city-scale crowd of millions of people**
- Accessing **one central place** within only a few hours

> **Caution**
> **Uncoordinated** access within confined area can **escalate into crowd disasters**!

## The Way to the Jamarat Bridge

**Question:** Where on the way does a crowd become dangerous?

## Crowd Accidents

- **Historical incidents** of crowd disasters
- Resulted in **casualties and injuries**
- **High-density bottlenecks** on the way to the rituals
- Several **critical points of congestion**
- Waiting times can lead to **hazardous conditions**

**Question: What could we do to prevent this?**

# Pedestrian Traffic

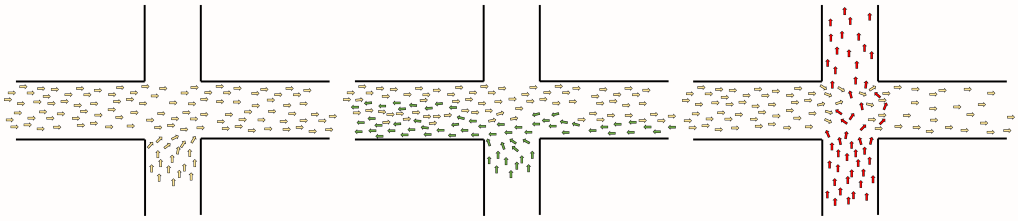
How is Hajj pedestrian traffic different from the regular urban pedestrian traffic in cities?

## Pedestrian Traffic

- Individuals and groups
- Multitude of destinations
- Mixing and formation
- Distractions

- Homogeneous groups
- Shared destination
- Higher densities
- Predictable

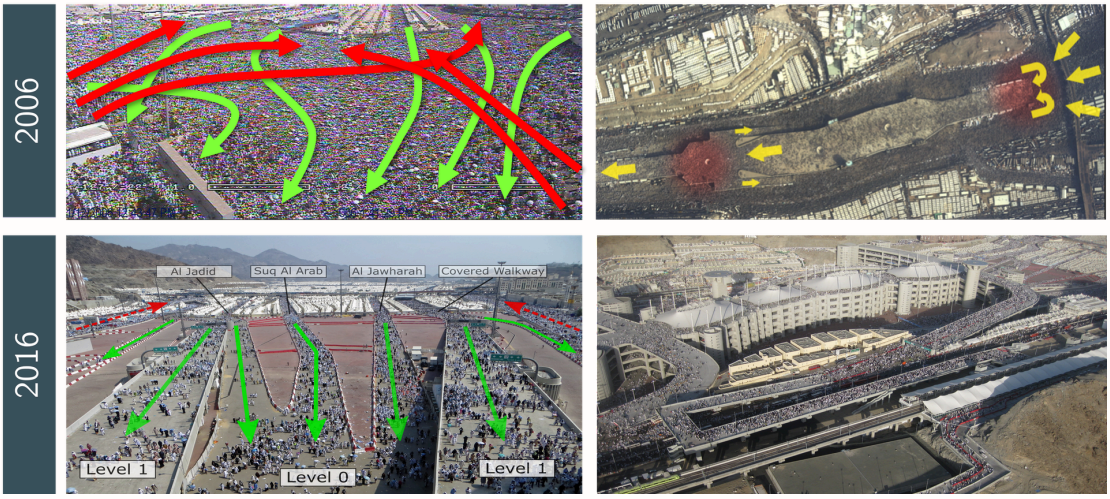
## Types of Pedestrian Flow



**Question: What is the most dangerous type here?**

> **Caution**
> **Multi-directional and intersecting flows** are the most dangerous type!

## Pilgrim Flows




> **General idea**
> Adhere to **one-way flow systems** and define path options for each camp under consideration of a unidirectional flow system.

# Problem Structure

## Objective?

**Question: What could be the objective?**

- Minimize **risk of overcrowding and accidents**
- **Enable ritual participation** of all pilgrims
- **Satisfy time preferences** of pilgrims
- **Easy plans to execute** under pressure

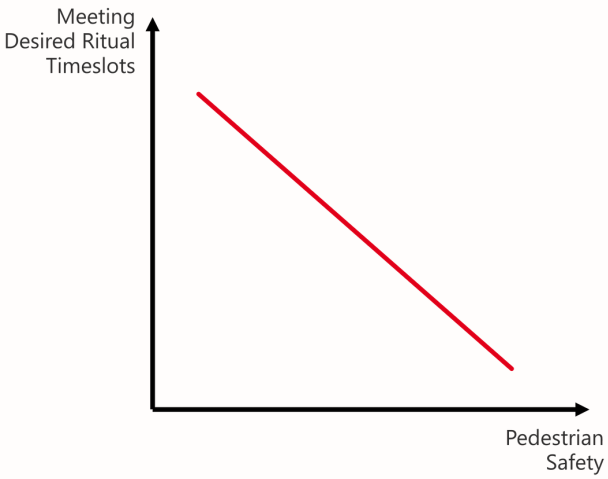
**Question: How can we try to model this?**

## Objective

- Satisfy **time preferences as much as possible**
- Consideration of **infrastructure bottleneck flow capacities**
- **Maximize safety** for all pilgrims
- **Simple plans** to make the execution as simple as possible
- This can **prevent critical errors** later on!

**Question: Where is the goal conflict?**

## Goal Conflict

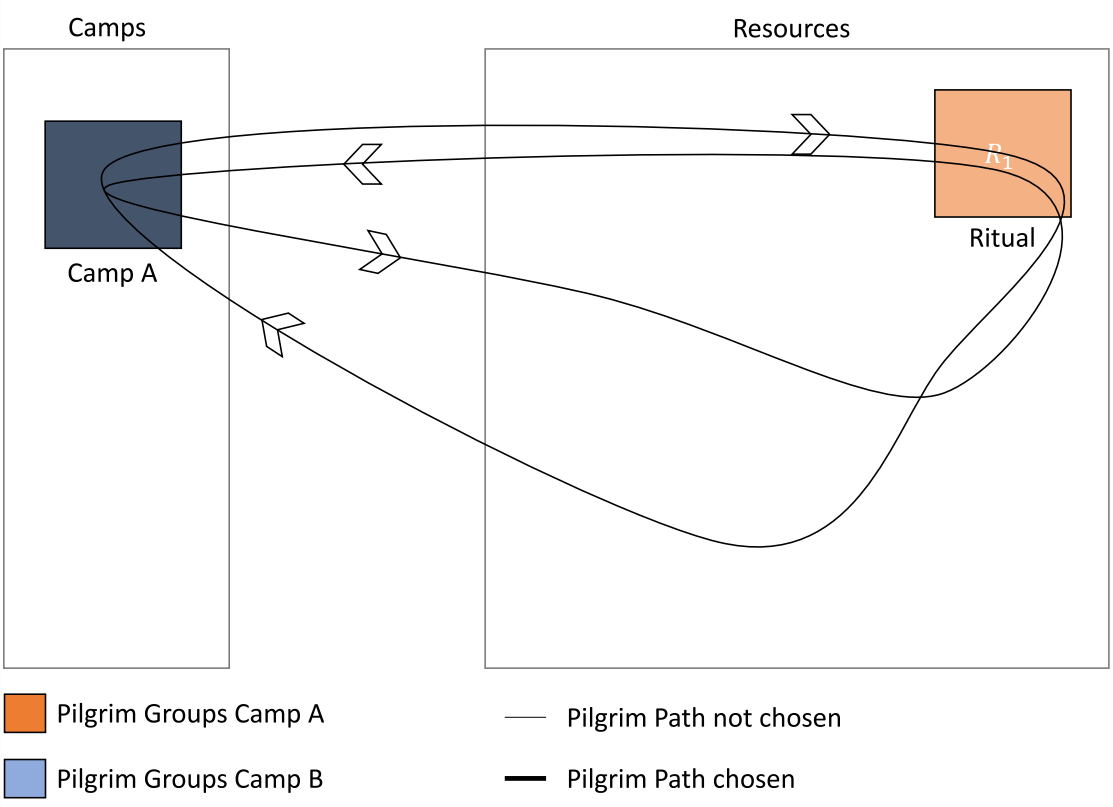



## Basic Structure

- We follow the structure of a **simple scheduling problem**
- The aim is to "assign" something to **different time periods**
- We assign **time slots to groups using different paths**
- Plans are kept simpler by assigning **paths to camps**
- Assignments are **fixed over the entire time horizon**

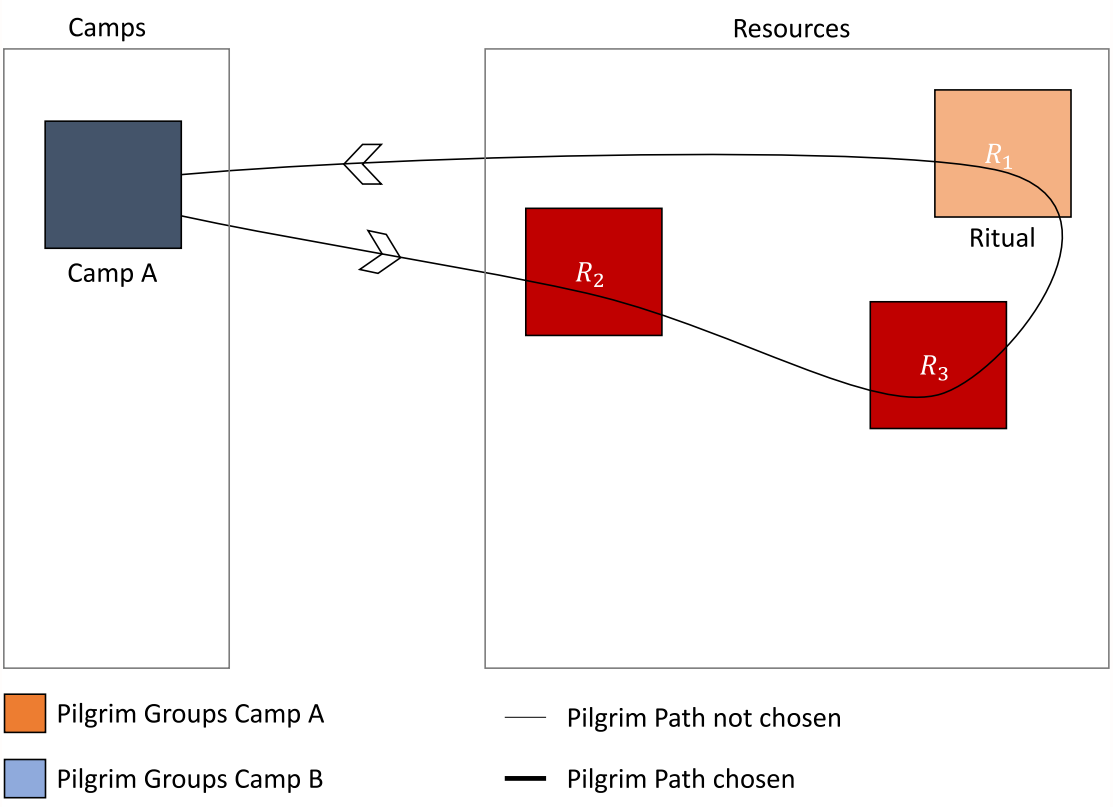
## Pilgrim Routes

Each camp has a set of feasible one-way paths that include the stoning ritual.



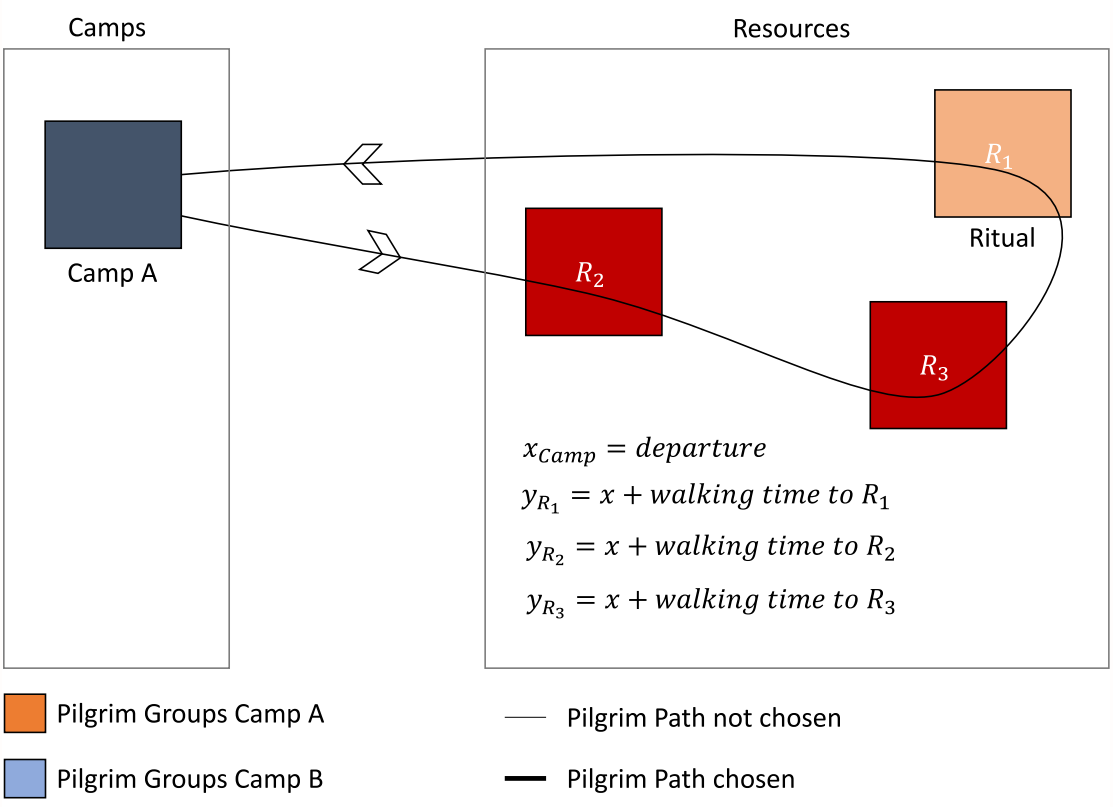

## Pilgrim Routes

A path may contain one or more bottlenecks, regarded as resources subject to a capacity.



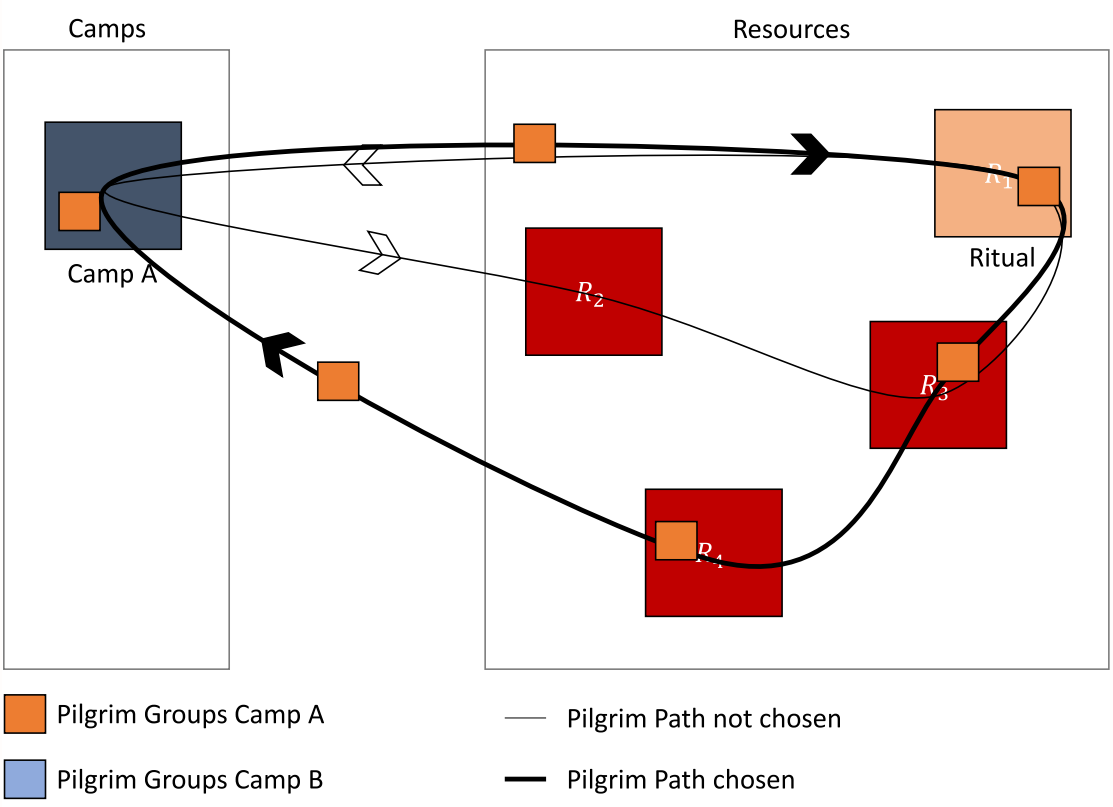

## Pilgrim Routes

Pilgrims depart from a camp at a time $t$ and pass through the bottleneck later.



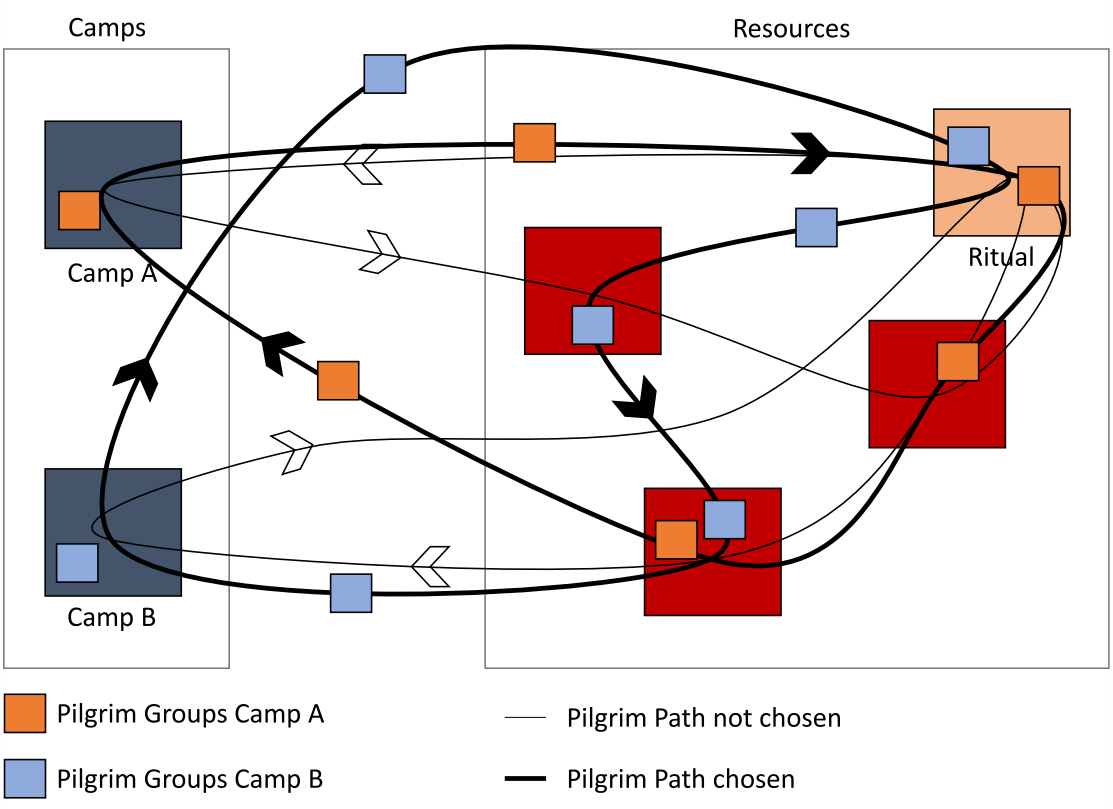

## Pilgrim Routes

Our model should assign one of the feasible paths to a camp on all four ritual days.



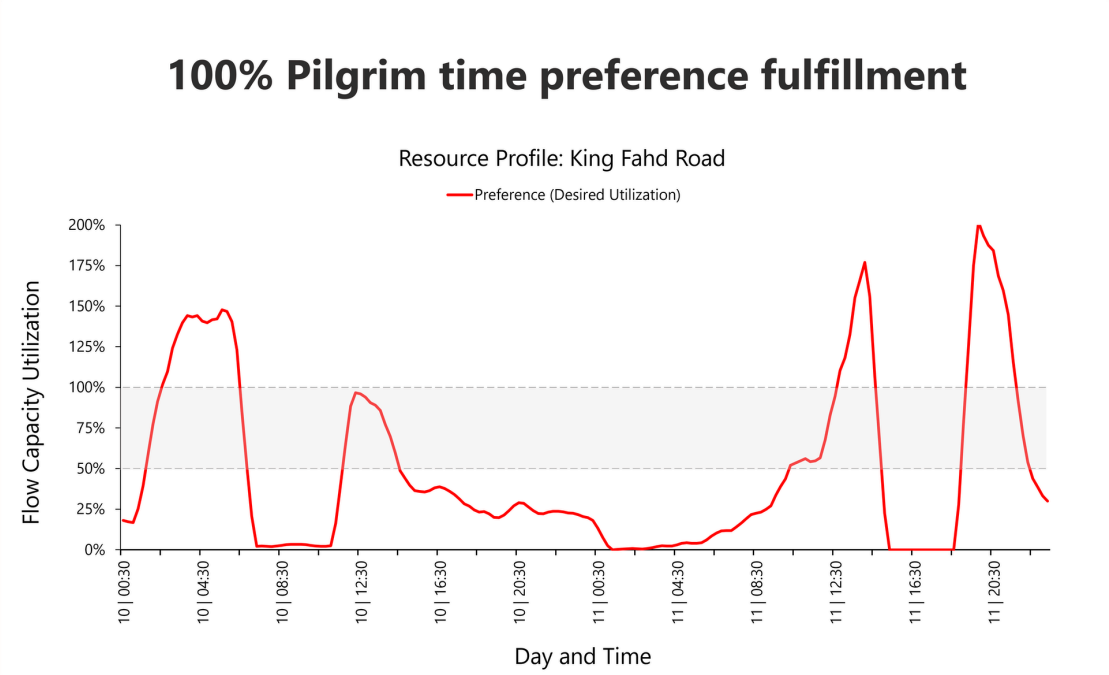

## Pilgrim Routes

These bottlenecks should not be overcrowded at any time during the Hajj.




How can we model
time preferences?

## Time Preferences

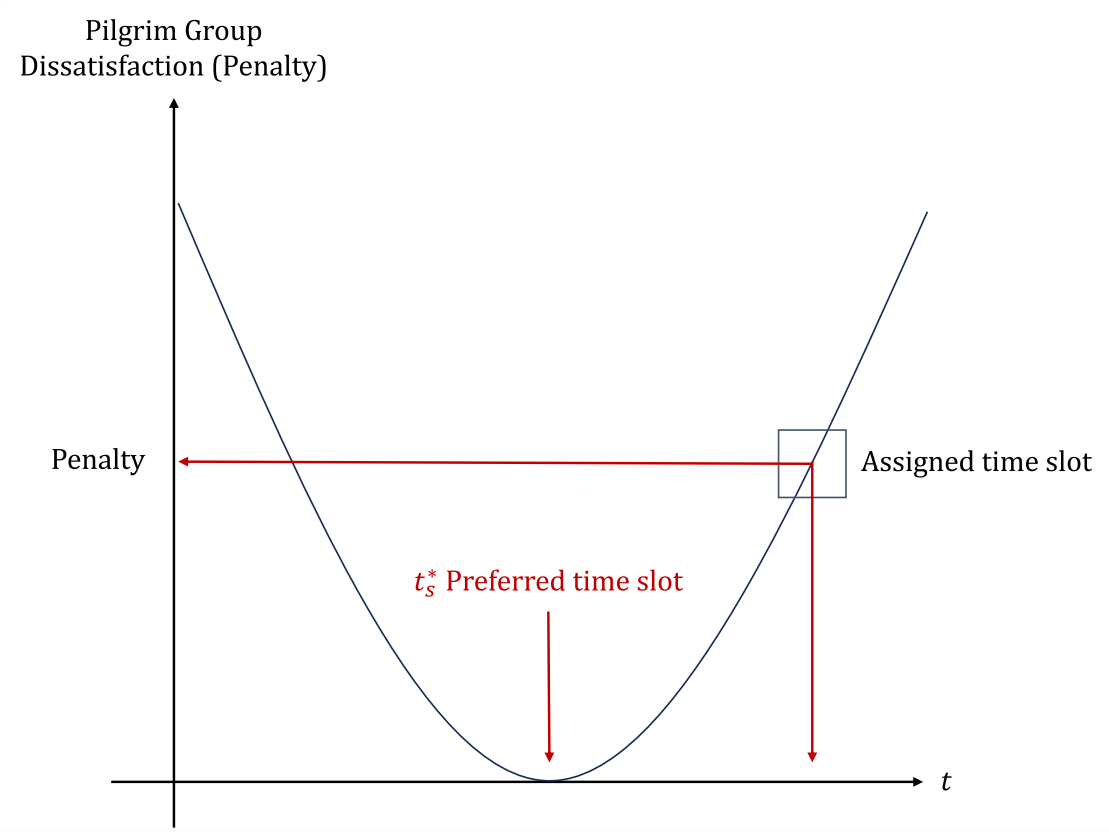



## Time preference satisfaction

- Assign **one departure time slot**
- Assigned per ritual day to each pilgrim group
- **Minimize difference between assigned and preferred time**
- Different **penalty functions** are possible

> **Group time preferences**
> May be computed, i.e., down-sampled given a distribution of pilgrims over time.

## Penalty Functions




## Fluctuations

**Question: What could become a problem?**

- If allowed demand between periods **varies strongly**, accidents are **more likely to happen!**
- We need to keep the **changes between periods within bounds**

**Question: Any idea how we can do that later?**

- Restrict the change of the utilization **between periods**

## Goals Summarized

1. Satisfy **time preferences of the pilgrims** as much as possible under the consideration of **infrastructure bottleneck flow capacities** by assigning “something” to a time slot.
2. For the sake of **simplicity and safety**, pilgrims coming from one camp will always have to be assigned **the same path**.
3. We need to keep track of the **relative utilization** of each resource **to restrict the fluctuations** between periods to ensure a safer event.

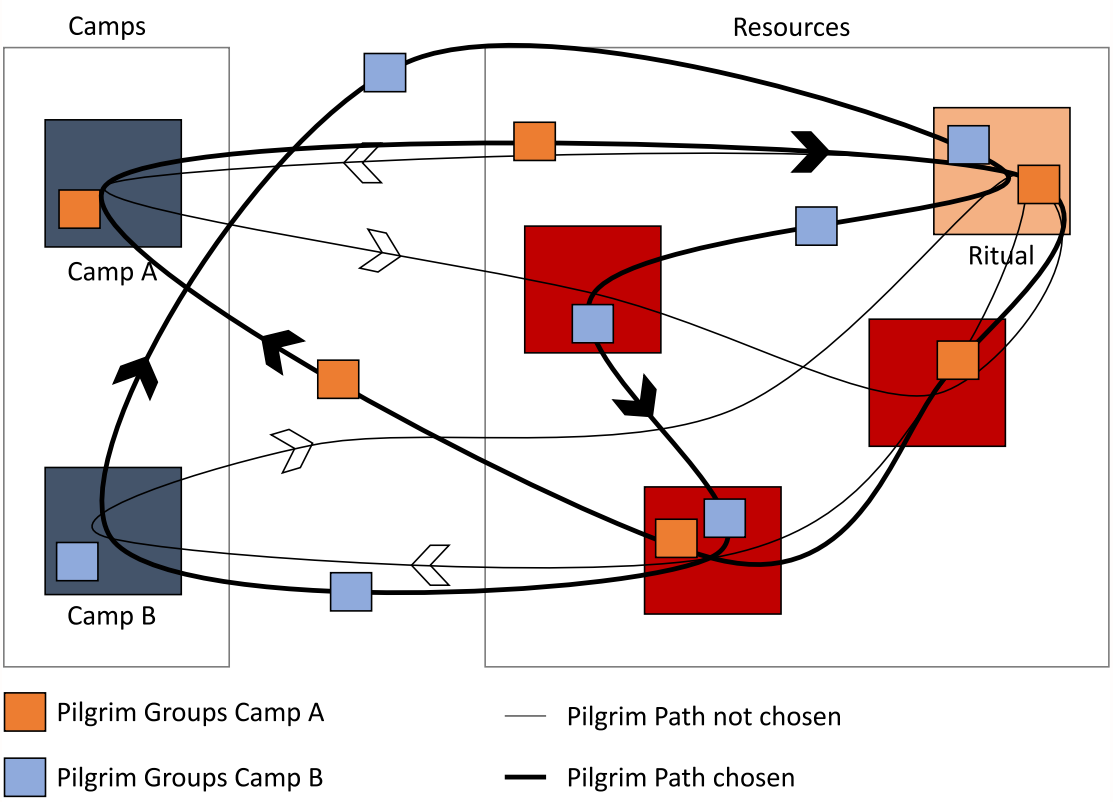
### Your Turn

Play: https://beyondsimulations.github.io/Applied-Optimization/games/pilgrims.html

# Model Formulation

### Sets?

**Question: What could be the sets here?**




### Sets

- $\mathcal{T}$ - Stoning periods in ascending order, indexed by $t$
- $\mathcal{R}$ - Infrastructure resources, indexed by $r$
- $\mathcal{C}$ - Pilgrim camps, indexed by $c$
- $\mathcal{P}$ - Paths that include the stoning, indexed by $p$
- $\mathcal{S}$ - Scheduling groups, indexed by $s$

> **Note**
> A **scheduling group** is a group of pilgrims from one camp on **one specific ritual day**. One model thus covers all four days at once.

**But we further need subsets!**

## Subsets

- $\mathcal{S}_c$ - Scheduling groups in camp $c$
- $\mathcal{S}_p$ - Scheduling groups that can use path $p$
- $\mathcal{P}_c$ - Feasible paths for camp $c$
- $\mathcal{P}_s$ - Feasible paths for group $s$
- $\mathcal{P}_r$ - Paths that contain the resource $r$
- $\mathcal{T}_s$ - Available stoning periods for group $s$ on its ritual day

---

That looks

complicated...

## On Subsets

**Question: Why use subsets?**

- It may seem **like a lot**
- But it also **really helps a lot!**
- We **reduce the problem size**

> **Tip**
> A smaller problem size **reduces the solution space** and **helps the solver** in finding the optimal solution faster!

## Parameters?

**Question: What could be possible parameters?**

- $n_s$ - Number of pilgrims in scheduling group $s$
- $f_{s,t}$ - Penalty value of assigning period $t$ to group $s$
- $a_{p,r}$ - Offset between stoning and utilization period of $r$ on $p$
- $b_{r,t}$ - Capacity of resource $r$ in period $t$
- $\sigma_r$ - max. change in relative utilization of $r$ between periods

## First Decision Variable?

> **Our first goal is to:**
> Satisfy time preferences of the pilgrims as much as possible under the consideration of infrastructure bottleneck flow capacities by assigning “something” to a time slot.

> **We need the following sets:**
> - Scheduling groups, $s \in \mathcal{S}$
> - Stoning periods in ascending order, $t \in \mathcal{T}$
> - Paths that include the stoning of the devil, $p \in \mathcal{P}$

## First Decision Variable

**Question: What could be our decision variable?**

- $X_{s,t,p}$ - 1, if scheduling group $s$ is scheduled to perform stoning in period $t$ and to use path $p$, 0 otherwise.

**Question: Do you get the idea here?**

It's a **binary assignment** of a group to a time slot and a path.

## Second Decision Variable?

> ! **Our second goal (more a constraint):**
> For the sake of **simplicity and safety**, pilgrims coming from one camp will always have to be assigned the same path.

> i **We need the following sets:**
> - Pilgrim camps from which groups can depart, $c \in \mathcal{C}$
> - Paths that include the stoning of the devil, $p \in \mathcal{P}$

**Question: What could be our second variable?**

## Second Decision Variable

- $Y_{c,p}$ - 1, if camp $c$ is assigned to use path $p$, 0 otherwise

**Question: Does anyone remember the third part?**

## Third Decision Variable?

> ! **Our third goal (again, more a constraint):**
> We need to keep track of the relative utilization of each resource to restrict the fluctuations between periods to ensure a safer event.

> i **We need the following sets:**
> - Infrastructure resources, $r \in \mathcal{R}$
> - Stoning periods in ascending order, $t \in \mathcal{T}$

**Question: What could be our third variable?**

## Third Decision Variable

- $U_{r,t}$ - Relative utilization of $r$ in $t$ with $0 \leq U_{r,t} \leq 1$

**Question: What does relative utilization mean?**

- It's a **percentage** of the capacity usage of the resource
- **Normalizes** the capacities between different resources

Let's start with our

objective function!

## Objective Function?

> ! **Our main objective is to:**
> Satisfy time preferences of the pilgrims as much as possible under the consideration of infrastructure bottleneck flow capacities by assigning "something" to a time slot.
> **Hint:** We thus could aim to minimize the total dissatisfaction with the timetable.

**Question: How could we minimize the total dissatisfaction?**

- **Penalize** difference between assigned and preferred time
- Different penalty functions, e.g., linear, quadratic, etc.

## Objective Function

> i **We need the following parameters and variables:**
> - $f_{s,t}$ - Penalty value of assigning period $t$ to group $s$
> - $X_{s,t,p}$ - 1 if group $s$ is scheduled to perform stoning in $t$ and to use $p$, 0 otherwise

**Question: What could be our objective function?**

$$\text{Minimize} \quad \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}_s} \sum_{p \in \mathcal{P}_s} f_{s,t} \times X_{s,t,p}$$

Note how we use the subsets $\mathcal{T}_s$ and $\mathcal{P}_s$ **to keep the model small**.

## Objective Function Characteristics

**Question: Is our objective function linear?**

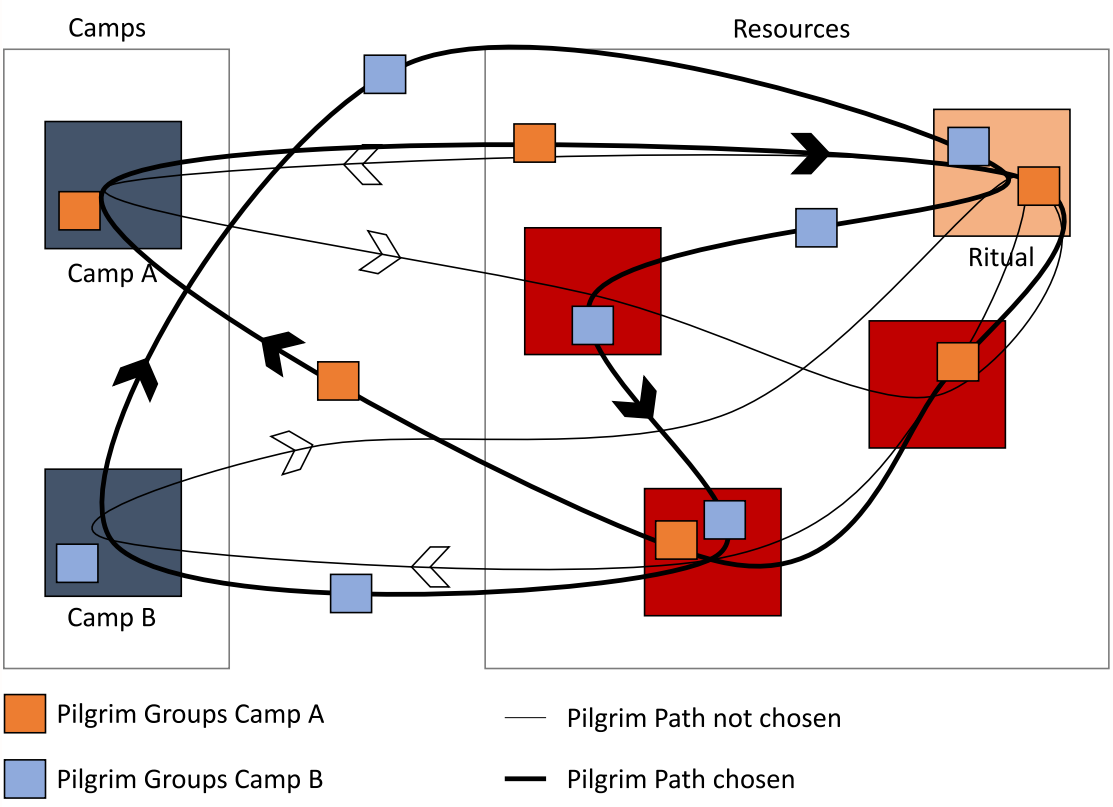
- We can use **non-linear penalty functions**
- But still, it will always be **linear**

**Question: Any idea why?**

- We can compute the **penalties in advance**
- Do **not** depend on the decision variables

# Constraints

## Constraints needed?




## Key Constraints

**Question: Which constraints do we need?**

1. Each group must have **one path assigned**
2. Each camp must have **one path assigned**
3. Each group must have **one time slot assigned**
4. Each resource must have **a capacity limit**
5. Constrain the **relative utilization between periods**

## Assign Paths to Camps

! **The goal of this constraint is to:**
Assign **one path to each camp** over the entire time horizon.

i **We need the following variables:**
- $Y_{c,p}$ - 1 if camp $c$ is assigned to use path $p$, 0 otherwise

**Question: What could be the constraint?**

$$\sum_{p \in \mathcal{P}_c} Y_{c,p} = 1 \quad \forall c \in \mathcal{C}$$

## Assign Time Slots to Groups?

! **The goal of this constraint is to:**
Assign **one time slot to each group** over the entire time horizon **using the same path** we have assigned to the camp in the previous constraint.

i **We need the following variables:**
- $X_{s,t,p}$ - 1 if group $s$ is scheduled to perform stoning in $t$ and to use $p$, 0 otherwise
- $Y_{c,p}$ - 1 if camp $c$ is assigned to use path $p$, 0 otherwise

**Question: What could be the constraint?**

## Assign Time Slots to Groups

$$\sum_{t \in \mathcal{T}_s} X_{s,t,p} = Y_{c,p} \quad \forall c \in \mathcal{C}, p \in \mathcal{P}_c, s \in \mathcal{S}_c$$

i **We use the following sets:**
- $\mathcal{C}$ - Pilgrim camps
- $\mathcal{S}_c$ - Scheduling groups in camp $c$
- $\mathcal{T}_s$ - Available stoning periods for scheduling group $s$
- $\mathcal{P}_c$ - Feasible paths for camp $c$

Each group can use all feasible paths of its camp: $\mathcal{P}_s = \mathcal{P}_c$ for all $s \in \mathcal{S}_c$.

## One Path and Slot per Group?

**Question: We wanted one path and one time slot per group. Where are these two constraints?**

Sum the last constraint over all paths of a camp:

$$\sum_{p \in \mathcal{P}_c} \sum_{t \in \mathcal{T}_s} X_{s,t,p} = \sum_{p \in \mathcal{P}_c} Y_{c,p} = 1 \quad \forall c \in \mathcal{C}, s \in \mathcal{S}_c$$

- Each group thus gets exactly **one period and one path**
- Some constraints can **imply** other constraints!

## Relative Utilization and Capacities

! **The goal of this constraint is to:**
Compute the relative utilization of each resource while also ensuring that the utilization does not exceed the capacity limit. **This one is very tricky!**

**Difficulties:**

- Includes the time-shift between stoning and utilization
- Used as parameter to shift periods in variable $X_{s,t,p}$

## Compute Relative Utilization?

> **We need the following:**
> - $n_s$ - Number of pilgrims in scheduling group $s$
> - $a_{p,r}$ - Period offset between stoning period and utilization period of $r$ on $p$
> - $b_{r,t}$ - Capacity of resource $r$ in period $t$ in number of pilgrims
> - $X_{s,t,p}$ - 1, if $s$ is scheduled to perform stoning in $t$ and to use $p$, 0 else
> - $U_{r,t}$ - Relative utilization of resource $r$ in period $t$ with $0 \leq U_{r,t} \leq 1$

**Question: What could be the constraint?**

## Compute Relative Utilization

$$\sum_{p \in \mathcal{P}_r} \sum_{s \in \mathcal{S}_p} n_s \times X_{s,t-a_{p,r},p} = b_{r,t} \times U_{r,t} \quad \forall r \in \mathcal{R}, t \in \mathcal{T}$$

> **Convention**
> Terms where the shifted period $t - a_{p,r}$ lies outside of $\mathcal{T}_s$ are simply dropped, as no group can be scheduled for stoning there.

## Where is the Capacity Limit?

**Question: We promised a capacity limit, but there is no "$\leq$ capacity" constraint. Why does it still hold?**

- It is **hidden in the variable bound** $U_{r,t} \leq 1$
- The equality forces the flow to equal $b_{r,t} \times U_{r,t}$
- Hence, the flow on $r$ can never exceed the capacity $b_{r,t}$

> **Tip**
> A variable domain and a constraint can **combine** into a constraint that is never written down explicitly.

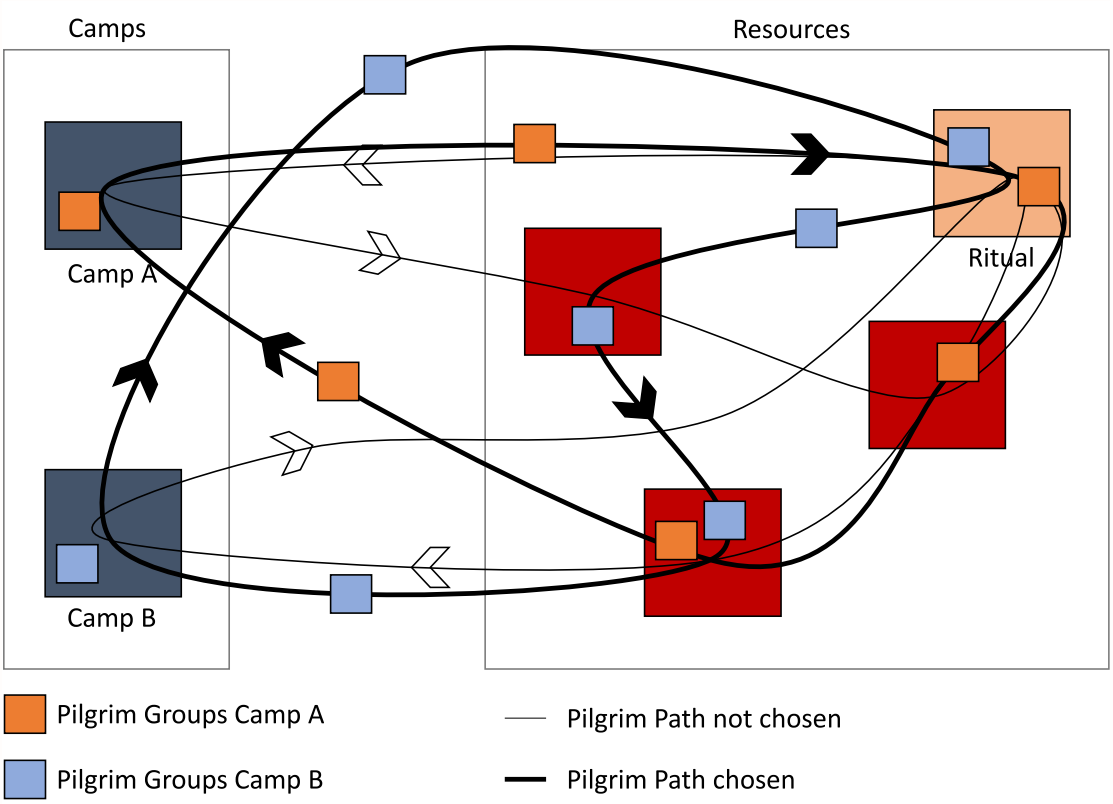
---

Let's pause!

Have you understood

this part?

## How does the shift work?




## Fluctuation Bounds?

! **The goal of this constraint is to:**
Keep the relative utilization of each resource within bounds to ensure a safer event.

i **We need the following:**
- $\sigma_r$ - max. change in relative utilization of $r$ between periods
- $U_{r,t}$ - Relative utilization of resource $r$ in period $t$ with $0 \leq U_{r,t} \leq 1$

**Question: What could be the constraint?**

## Fluctuation Bounds

$$U_{r,t} - U_{r,t-1} \leq \sigma_r \quad \forall r \in \mathcal{R}, t \in \mathcal{T} : t > 1$$

$$U_{r,t-1} - U_{r,t} \leq \sigma_r \quad \forall r \in \mathcal{R}, t \in \mathcal{T} : t > 1$$

## Why Does This Work?

**Question: Can somebody explain why this works?**

- Each constraint limits the **change**
- The first one limits the **increase**

- The second one limits the **decrease**
- We start at $t > 1$, as no period exists before the first one

### Scheduling Problem I

subject to:

### Scheduling Problem II

> **Note**
> Restricting the **relative utilization** of each resource to a certain bound.

### Scheduling Problem III

> **Note**
> All variables, except for $U_{r,t}$, are binary.

## Model Characteristics

### Characteristics

**Questions: On model characteristics**

- Is the model formulation linear/ non-linear?
- What kind of variable domains do we have?
- Have we specified the length of a period?

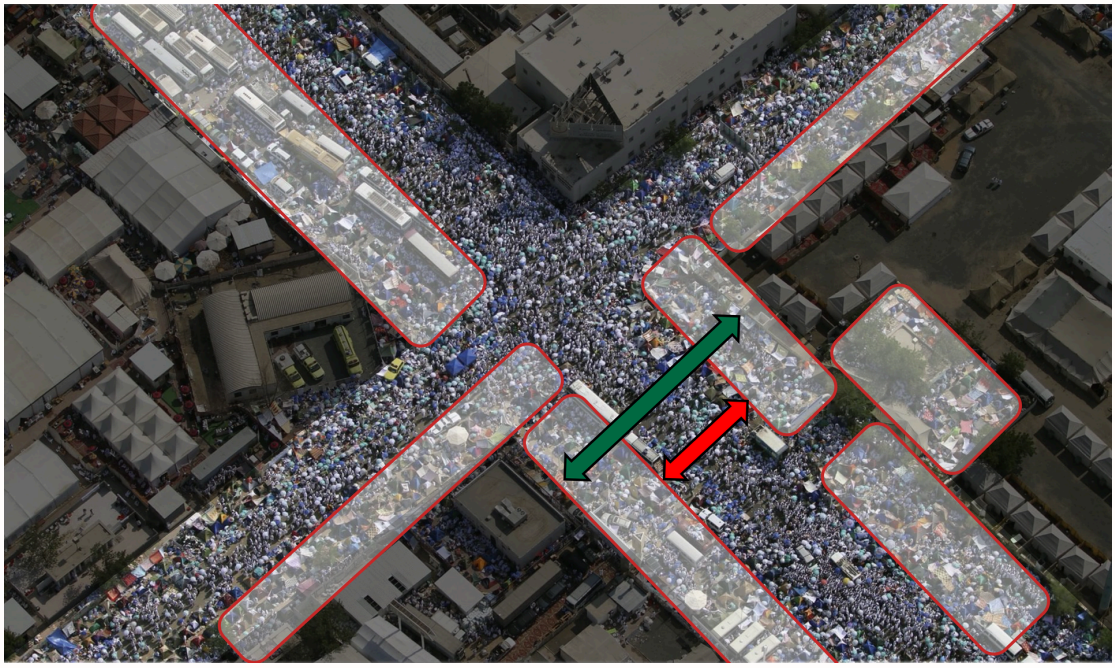
### Model Assumptions

**Questions: On model assumptions**

- What assumptions have we made?
- What are likely issues that can arise if applied?
- How can we measure flow capacities?
- Are all pilgrims equally fast?

> **Important**
> Capacities are **hard limits**: if demand exceeds them, the model becomes **infeasible** and refuses to produce an unsafe plan. This is a feature, not a bug!

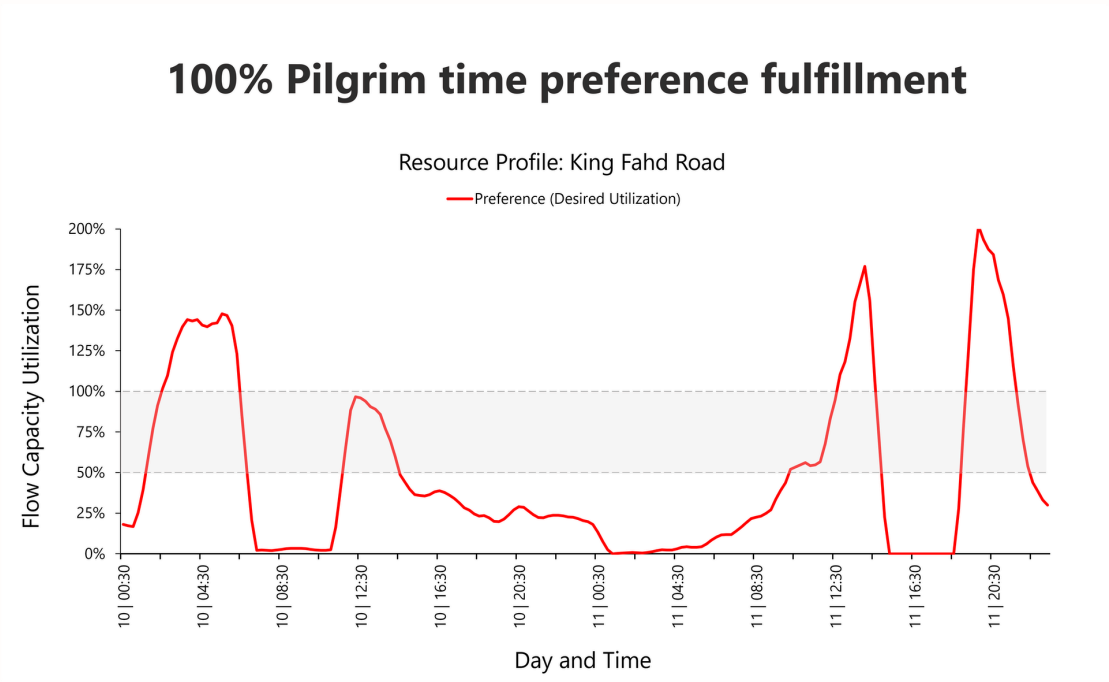
## Capacity Buffers

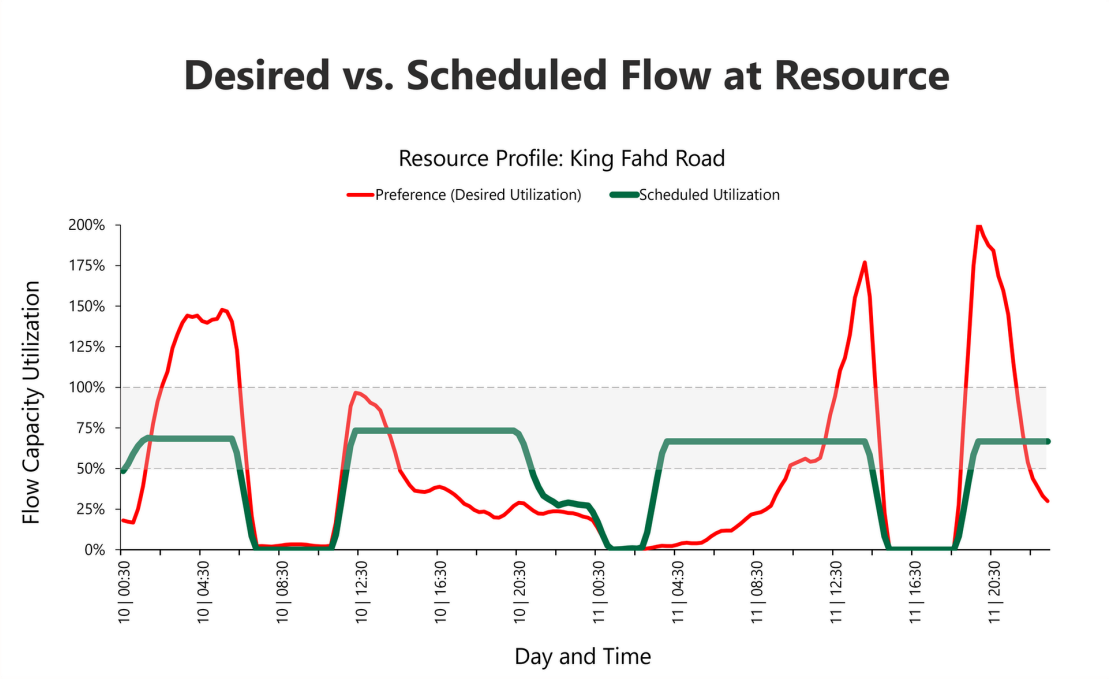


# In Practice

Can this be

applied?



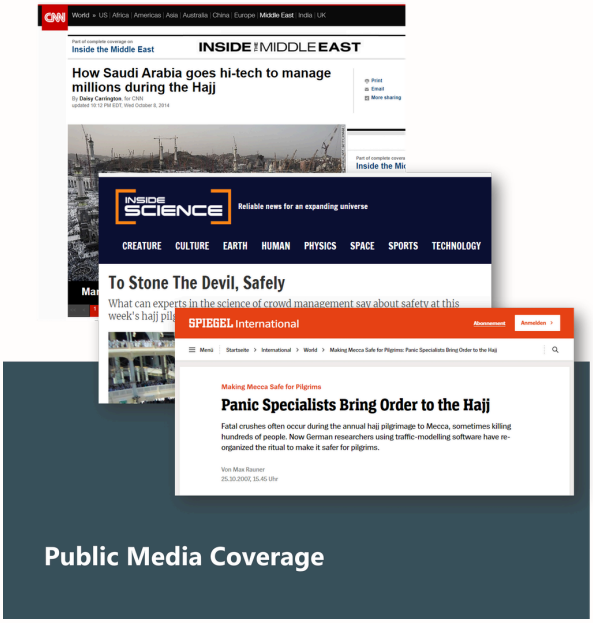




## Implementation

- Optimization **part of a bigger picture**
- Many projects with several disciplines involved

- E.g. Simulations, infrastructure projects, real-time monitoring, contingency plans, awareness campaigns, ...

> **Note**
> Optimization was part of a project by Knut Haase and his team [1].



His Excellency, Dep. Minister Habib Zain Al-Abideen presenting at informs Analytics 2015

## Wrap Up

> **And that's it for today's lecture!**
> We now have covered a scheduling problem based on a real-world application and are ready to start solving some new tasks in the upcoming tutorial.

Questions?

# Literature

To learn more about the project and crowd management during the Hajj, take a look at K. Haase *et al.* [1] and the literature list of this course.

[1] K. Haase *et al.*, "Improving pilgrim safety during the hajj: an analytical and operational research approach," *Interfaces*, vol. 46, no. 1, pp. 74–90, 2016.